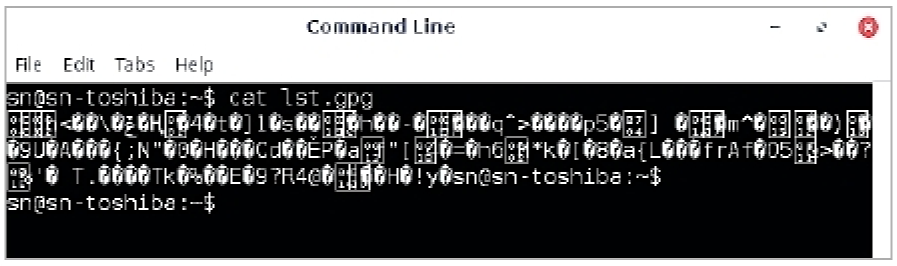

Figure 4: Display of an encrypted file using the *cat* command

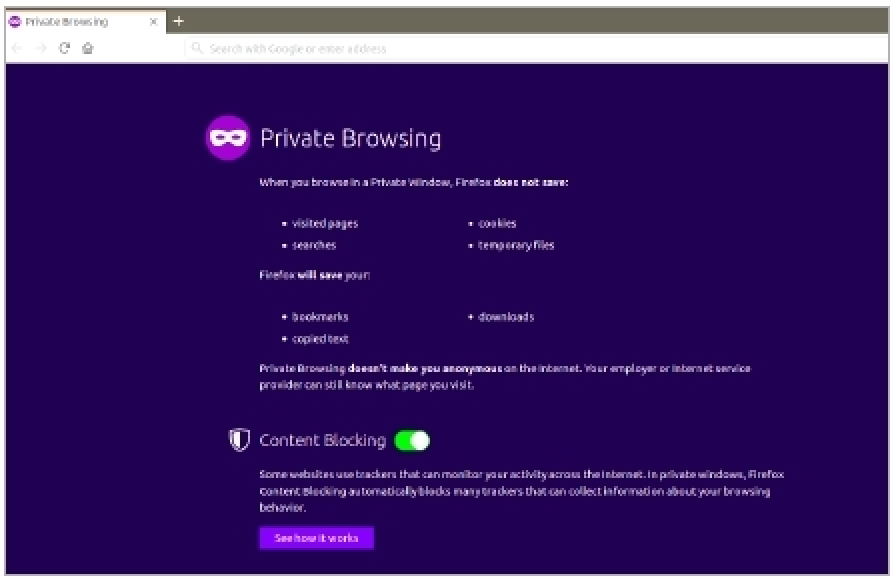

Figure 5: Firefox mode in private/incognito mode

in an easy way, one can try GnuPG2, an easy-to-use open source tool, which is a free implementation of the OpenPGP standard. It permits encryption with symmetric as well as asymmetric methods. As we are considering encryption for personal use, I will describe the symmetric method that uses the same passphrase for both encryption as well as decryption without the need to exchange keys. The utility is available on most distributions of Linux and Windows.

Install GnuPG2 with the following command on an Ubuntu system:

```
$sudo apt-get install gnupg2
```

The command to encrypt a file is as follows:

```
$gpg2 -symmetric trial.txt
```

The above command will ask for a passphrase and then create a new file called *trial.txt.gpg*, after which the original file can be deleted using *shred*.

The default algorithm used for encryption is AES128. If you want to use the AES256 cipher instead, you can specify it in the command in the following way:

```
$gpg2 -c --cipher-algo AES256 trial.txt
```

The encrypted file can then be decrypted by running the command *gpg2* with the name of the encrypted file as the parameter. The passphrase must be entered, following which the command will produce the decrypted file which is the same, but with the *.gpg* extension removed from the file name.

```
$gpg2 trial.txt.gpg
```

The *-o* option can be used to give the name of the output file, both for encryption and decryption.

```
$gpg -o trial.txt.gpg trial.enc
$gpg -o trial.dec trial.enc
```

Note: The password for decryption will not be asked for if you use gpg2 to decrypt the file within a certain time, the duration of which can be configured as per your requirements.

Browser related privacy issues: Clearing the browser cache, cookies and history

The browser caches the recent sites visited, logs the sites visited and stores the cookies (if any) of the websites visited. Such data can be cleared to protect privacy and confidentiality.

In Chrome, the browser cache, cookies and the history can be cleared by pressing the *Ctrl-Shift-Del* keys. Choose the time range to clear the cache. The *All time* option clears the cache and history completely.

In Firefox, press *Ctrl-Shift-Del* and choose the time range and the categories of data that you want to delete.

Browser settings for privacy

Opening browsers in private mode: Most browsers have a private mode, which can be used to eliminate any traces of the sites visited in the local system. Hence, all traces of usage are eliminated once the browsing window is closed. This is particularly useful in shared systems.

In Chrome, the browser can be started in private mode by using *Ctrl-Shift-N* and is called the incognito mode.

In Firefox, the browser can be started in private mode by using *Ctrl-Shift-P* and is called the private window.

Tracking protection: Many websites do cross-site tracking by saving cookies in the user's system to learn about the websites visited, search subjects, etc. This can be prevented by turning off *Browser tracking* in the browser settings.

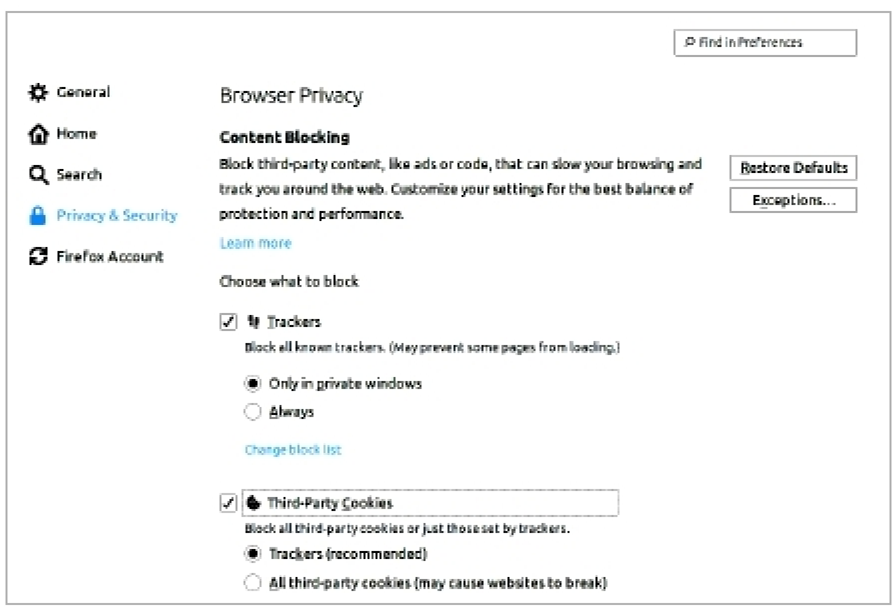

Figure 6: Tracking protection in Firefox